

Requirements Document

1.0 Introduction/Overview

Title: Preventive Maintenance Service for Thermo Orbitrap Mass Spectrometer Systems

1.1 Background

The Clinical Cancer Metabolism Facility at the National Institutes of Health (NIH), National Cancer Institute (NCI), Center for Cancer Research (CCR), utilizes a Thermo Orbitrap Lumos mass spectrometer system and Thermo Orbitrap Exploris 240 mass spectrometer system for time-sensitive metabolic analyses of clinical tissue samples to support treatment of patients enrolled in clinical trials conducted at the NIH Clinical Center. The CCR Mass Spectrometry Resource, NCI, uses a Thermo Orbitrap Lumos mass spectrometer system for proteomic analysis of cell and tissue samples to research the dysregulation that occurs in cancer cells and effects of treatments. The mass spectrometer systems are heavily used and must be maintained to ensure peak functionality and maximum productivity.

1.2 Scope

The Contractor shall provide all labor, material and equipment necessary to maintain and provide preventive maintenance for the following equipment system:

Item	Description	Catalog #	Serial #
1	Thermo Orbitrap Fusion Lumos mass spectrometer	FSN02-10001	FSN20703
2	Thermo Dionex ISC-6000 DP dual pump	22181-60009	20056143 and 200096396
3	Thermo Dionex ICS-6000 EG eluent generator	22181-60019	20066083
4	Thermo Dionex AS-AP autosampler with temperature control	074926	20073003
5	Thermo ICS-6000 Detector/Chromatography (DC) Microbore Compartment with dual temperature zones and two injection valves	22181-60049	20056143
6	Thermo Ion Chromatography	079829	20159198

7	Thermo Ion Chromatography CD Conductivity Detector	079829	20269275
8	Thermo Orbitrap Exploris 240	BRE725535	MM10734C
9	Thermo Dionex U3000 Binary Pump	VH-P10-A-02	8333786 and 8333800
10	Thermo Dionex U3000 Column Compartment	VH-C10-A-03	6510445 and 6510459
11	Thermo Dionex U3000 Diode Array Detector	VF-D11-A-01	8333550
12	Thermo Dionex U3000 Split Sampler	VH-A40-A-02	8333016
13	Thermo Orbitrap Fusion Lumos mass spectrometer	FSN02-10001	FSN20265
14	Thermo Dionex U3000 RSLCnano Autosampler	5826.0020	8121502
15	Thermo Dionex U3000 RSLCnano Nanopump	5041.0010A	8120696
16	Thermo Dionex U3000 RSLCnano Column Compartment	TCC-3000RS	8120696

All maintenance services shall be performed on-site in accordance with the most current standard commercial maintenance practices. Onsite and emergency services shall be provided during standard business hours (defined as 8:00 am – 5:00 pm ET, Monday through Friday, excluding Federal Holidays found at www.opm.gov/policy-data-oversight/pay-leave/federal-holidays and extended hours (5:00 pm – 9:00pm ET, Monday through Friday) for any issues not able to be resolved remotely, or during normal business hours. Software updates/services may be provided remotely as applicable. All travel, labor, materials, and related charges shall be included in the purchase order price.

1.3 Objectives

The objectives of this effort are to:

- Ensure the continuous, reliable, and optimal operation of Thermo Orbitrap mass spectrometer systems and associated components
- Minimize system downtime through preventive maintenance and timely emergency

repair services

- Maintain instrument performance in accordance with manufacturer specifications to support accurate, reproducible, and time-sensitive clinical and research activities at NIH

1.4 Type of Contract

This is a firm-fixed-price purchase order.

1.5 Consideration and Payment

In consideration of satisfactory performance of the work as described throughout this contract, the Contractor shall be paid a monthly fixed price for the services identified below. Payment shall be made monthly in arrears upon the delivery and acceptance of the services required.

Item	Description	Catalog #	Serial #	Price
1	Thermo Orbitrap Fusion Lumos mass spectrometer	FSN02-10001	FSN20703	\$TBD
2	Thermo Dionex ISC-6000 DP dual pump	22181-60009	20056143 and 200096396	\$TBD
3	Thermo Dionex ICS-6000 EG eluent generator	22181-60019	20066083	\$TBD
4	Thermo Dionex AS-AP autosampler with temperature control	074926	20073003	\$TBD
5	Thermo ICS-6000 Detector/Chromatography (DC) Microbore Compartment with dual temperature zones and two injection valves	22181-60049	20056143	\$TBD
6	Thermo Ion Chromatography	079829	20159198	\$TBD
7	Thermo Ion Chromatography CD Conductivity Detector	079829	20269275	\$TBD
8	Thermo Orbitrap Exploris 240	BRE725535	MM10734C	\$TBD

9	Thermo U3000 Binary Pump	VH-P10-A-02	8333786 and 8333800	\$TBD
10	Thermo U3000 Column Compartment	VH-C10-A-03	6510445 and 6510459	\$TBD
11	Thermo U3000 Diode Array Detector	VF-D11-A-01	8333550	\$TBD
12	Thermo U3000 Split Sampler	VH-A40-A-02	8333016	\$TBD
13	Thermo Orbitrap Fusion Lumos mass spectrometer	FSN02-10001	FSN20265	\$TBD
14	Thermo U3000 RSLCnano Autosampler	5826.0020	8121502	\$TBD
15	Thermo U3000 RSLCnano Nanopump	5041.0010A	8120696	\$TBD
16	Thermo U3000 RSLCnano Column Compartment	TCC-3000RS	8120696	\$TBD
			Total	\$TBD

2.0 Requirements

The Contractor shall provide all labor, materials, and equipment required to perform the following tasks:

2.1 Preventative Maintenance

The Contractor shall perform one (1) planned preventive maintenance visit during each 12-month performance period for each covered system. Technically qualified, , factory-trained service engineers with demonstrated expertise in the calibration, alignment, maintenance, and repair of Thermo Orbitrap Fusion Lumos and Orbitrap Exploris 240 systems and their integrated liquid chromatography components shall perform this service.

The required service shall consist of a thorough cleaning, calibration, adjusting, inspection, and testing of all equipment in accordance with the manufacturer's latest established service procedures. The Contractor shall perform calibration, alignment, verification and validation for each module of the instrument system according to manufacturer's latest established maintenance protocol. All equipment shall be operationally tested through at least one (1) complete operating cycle at the end of the preventive maintenance inspection to assure optimum and efficient performance. Preventive maintenance kits shall be included at no additional cost to the Government.

The required services include comprehensive preventive maintenance, emergency repair services, replacement of compatible parts (excluding consumables), software support, and unlimited remote technical assistance for Thermo Orbitrap Fusion Lumos and Orbitrap Exploris 240 mass spectrometer systems and their integrated Vanquish and Dionex chromatography modules. These highly specialized, mission-critical instruments are utilized for time-sensitive metabolic and proteomic analyses of clinical and research samples in support of NIH clinical trials and cancer research programs.

Fees for labor, travel, and preventative maintenance replacement components shall be included at no additional cost to the Government.

A report indicating the work performed shall be provided to NCI's Technical Point of Contact (TPOC) within three (3) business days of completion. The report shall be provided in electronic format (Word or PDF file).

2.2 Emergency Service

On-site, emergency repair service visit shall be provided by technicians with extensive experience servicing and repairing Orbitrap Fusion Lumos and Exploris series spectrometers as well as Vanquish and Dionex Chromatography systems during the term of this contract at no additional cost to the Government. Emergency service shall be provided during normal working hours, Monday through Friday excluding Federal Holidays (see Section 3.0). Upon receipt of complaint that any part of the equipment is not functioning properly, the Contractor shall provide an initial response within one (1) hour, acknowledging receipt of the complaint. If a complaint is made outside of the

business hours established in this section, the Contractor shall provide a response within one (1) hour of the start of the next business day.

The Contractor shall provide an on-site response within three (3) business days after receipt of the complaint. If a complaint is made outside of business hours, the Contractor shall provide an on-site response within three (3) business days from the start of the next business day.

A qualified, trained service representative with extensive experience servicing and repairing Orbitrap Fusion Lumos and Exploris series spectrometers as well as Vanquish and Dionex Chromatography systems shall be dispatched to inspect the equipment and perform all repairs and adjustments necessary to restore the equipment to normal and efficient operating condition. Emergency service calls shall not replace the necessity for planned preventative maintenance. All labor, travel, and compatible parts shall be included.

Within three (3) days of the completion of any emergency repair services performed, the Contractor shall provide a brief emergency service report to the TPOC stating the equipment malfunction that occurred, the repair efforts, and the duration that was required to restore the equipment to operation. The emergency service report shall be provided in electronic format (Word or PDF file).

2.3 Replacement Parts

The Contractor shall furnish all required replacement parts at no additional cost to the Government, except for consumables. All parts and components supplied under this agreement shall be fully compatible with the existing systems and equipment and capable of operating without modification to current configurations. Upon identification of required parts, they shall be delivered within 24 hours, or the following business day should the request fall on a Friday or preceding a Federal Holiday. If parts will take longer than 24 hours to be delivered for any reason, the Contractor must provide the TPOC with a notification stating the reason and estimated delivery timeframe immediately upon identification of delivery delay for approval by the Contracting Officer.

2.4 Updates/Service

The Contractor shall provide technical support for already-owned program software and troubleshooting of the operating systems, at no additional cost to the Government. The Contractor shall receive advance approval from the Government prior to installation of any software updates or revisions. Installation of any software shall be performed by qualified personnel.

The Contractor shall provide an electronic report to the TPOC per the within one (1) day after successful performance of software updates. This report shall also include the reason for the updates, and any effect it may have on equipment operation.

2.5 Technical Support

The Contractor shall provide unlimited remote technical support to the end-user by technicians with extensive experience servicing, repairing and troubleshooting Orbitrap Fusion Lumos and Exploris series spectrometers as well as Vanquish and Dionex Chromatography systems. Technical support shall include diagnosis of the issue encountered by the end-user. The Contractor shall provide diagnostic and remedial support services remotely through secure remote service tools to the greatest extent possible. If the issue can be corrected remotely, the Contractor shall perform remote corrective maintenance or walk the end-user through any simple adjustments and corrections necessary to resolve the issue. Technical support shall be provided by a certified service engineer or technically qualified, trained personnel.

The Contractor shall provide a toll-free telephone number for real-time remote technical support requests. The Contractor shall use its best efforts to respond to all technical support requests submitted via telephone immediately. If the Contractor is unable to respond immediately, the Contractor shall provide support within one (1) hour after the technical support request call is made by the end-user.

Real-time remote technical support via telephone shall be available during normal business hours, Monday – Friday, 8:00 a.m. – 5:00 p.m. Eastern Time, excluding Federal public holidays as defined by 5 U.S.C. 6103. Information on observed Federal holidays may be located at the OPM website: <https://www.opm.gov/policy-data-oversight/pay-leave/federal-holidays/>

In addition to the telephone number, the Contractor shall also provide an e-mail address and/or an online platform for technical support requests. Furthermore, the Contractor shall provide a web-based, self-help expert database for answering product questions, accessing documentation, and resolving issues.

If the Contractor cannot resolve a remote technical support request within three (3) business days, the Contractor shall provide on-site emergency repair services within three (3) business days from the last attempt at remote support in accordance with this Statement of Work.

2.6 Service Exclusions

The Contractor shall not be responsible for any repairs necessitated by abuse, neglect, vandalism, Acts of God, fire, or water. Should damage occur as a result of the aforementioned events, the Government reserves the right to obtain repairs needed as a result of such events via a modification to the purchase order, under a separate purchase order, or any other method deemed appropriate and in accordance with Federal Acquisition Regulations.

2.7 Personnel Qualifications

Only technically qualified trained personnel shall perform service on the Thermo Orbitrap Fusion Lumos equipment, Exploris 240 equipment at NIH, Dionex and Vanquish Chromatography systems, as covered by this purchase order. All primary service personnel shall have at least one backup support person with at least the same level of expertise on the equipment covered under this purchase order.

3.0 Deliverables

DELIVERABLE	REQUIRED FORMAT	DUE DATE
Preventive Maintenance Service Report (Section 4.1)	Electronic, Word or PDF format	3 business days after completion of service
Emergency Maintenance Service Report (Section 4.2)	Electronic, Word or PDF format	3 business days after completion of service

Software Updates/Revisions (Section 4.4)	Electronic, Word or PDF format	1 business days after completion of service
Delayed Delivery Notification (Section 4.3)	Electronic, Word or PDF format	Immediately upon identification of delayed deliveries

4.0 Period of Performance

The Period of Performance shall consist of one (1) 12-month base period and four (4) 12-month option periods as follows:

Base Period	April 30, 2026 – April 29, 2027
Option Period 1	April 30, 2027 – April 29, 2028
Option Period 2	April 30, 2028 – April 29, 2029
Option Period 3	April 30, 2029 – April 29, 2030
Option Period 4	April 30, 2030 – April 29, 2031

5.0 Place of Performance

Onsite services shall be performed at the following location:

NIH
Bldg. 10, Rm. 13N258
9000 Rockville Pike
Bethesda, MD 20892

NIH
Building 37, Room 2140
9000 Rockville Pike
Bethesda, MD 20892